

Sagar Mishra

Melbourne, Australia | sagarcroc@gmail.com | +61 424 308 228 | Full working rights in Australia
github.com/sagarmsraozil | sagarmsraozil.github.io | linkedin.com/in/sagar-mishra-a3455121b

Professional Summary

Software engineer with two years of commercial work at Programiz, a coding-education platform with over 100,000 paid learners. I built its course flow APIs and certificate engine in Node.js and TypeScript and integrated Paddle payments. During my Master's I led a team of three building a warehouse platform and a multi-tenant storefront in TypeScript on PostgreSQL, with a locking stock ledger and a separate payment service. At Jobss.ai in Melbourne I led four engineers.

Education

Victorian Institute of Technology, Melbourne, VIC

Feb 2024 – Jul 2026

Master of Information Technology and Systems (Software Engineering)

Softwarica College of IT and E-commerce, Kathmandu, Nepal

Nov 2018 – Oct 2021

BSc (Hons) Software Engineering

Skills

Languages & Frameworks: TypeScript, JavaScript, Node.js, Express, Kotlin, PHP, Laravel, Python, SQL, React, Next.js

Backend: REST APIs, payment integration, webhooks, authentication, role-based access control, idempotency, caching

Cloud Platforms: Vercel. Currently learning AWS (EC2, RDS, S3) and Docker

DevOps & Tools: Git, GitHub Actions CI, Playwright, Cypress, Postman, Sentry, Jira, Directus, Claude Code

Databases: PostgreSQL (Postgres), MySQL, MongoDB, Redis, Drizzle, Sequelize, row-level locking

Practices: Code review, Agile, spec-driven development, AI-assisted development

Professional Experience

Jobss.ai, Software Engineering Intern, Melbourne, VIC

Feb 2026 – May 2026

- Led four engineers on a Laravel and PHP job platform, live in production. Reviewed every pull request before it merged.
- Owned authentication and role-based access control in Laravel for four roles, all checked against one permission model.
- Designed the employer workspace in Laravel and Blade. Applicants move through review, shortlist and rejection.
- Built marketplace search on one index that serves both sides, people looking for work and people looking to hire.

ParewaLabs (Programiz), Software Engineer, Kathmandu, Nepal

Oct 2021 – Nov 2023

- Built the course flow APIs in Node.js, TypeScript and MySQL. Over 100,000 paid learners use them to finish a course.
- Integrated the Paddle payment API, which every paid learner pays through. Wrote the webhooks and the checkout iframe messaging. Programiz still takes payment through that workflow today.
- Built the certificate engine in Node.js and TypeScript. It has issued over 100,000 verified completion certificates.
- Owned the online compiler, which drew around 23 million visits a month from Google search. Fixed Python output arriving over the socket in pieces with a heartbeat that joins it into one response, working with the principal engineer.
- Led a team of six on Programiz Reports. Took it from content-team requirements to release in two weeks.
- Integrated sensAI, an LLM-backed tutor, inside the exercise page. Learners ask it over 300 questions a day.
- Wrote end-to-end tests in Playwright and Cypress and ran them in a GitHub Actions CI workflow.

Projects

Nexus and Aroma were built by a team of three that I led. I architected both, wrote the spikes and specs, and reviewed every pull request, over 400 in two years. AI models implemented against the frozen specs.

Nexus, warehouse management platform. TypeScript, Node.js, PostgreSQL, Redis, Next.js

Feb 2025 – Jan 2026

- Designed the stock ledger in PostgreSQL with SELECT FOR UPDATE row locks, so two buyers cannot claim one unit.
- Designed the fulfilment pipeline. Cron jobs batch retail orders, and each order is tracked as its own state machine.
- Modelled product drops as a state machine through supplier onboarding, lot negotiation, intake and pricing.
- Now rebuilding it as e-warehouse with an agentic workflow in Claude Code. I write the spikes, assumptions and some of the code. Agents implement the rest, and I review their code before it merges.

Aroma, multi-tenant eCommerce storefront. TypeScript, Node.js, PostgreSQL, Redis, Next.js

Mar 2024 – Jan 2026

- Designed the tenancy model so each seller runs an isolated storefront on one shared backend.
- Specified checkout to reserve stock on order confirmation. An idempotency key stops a retry creating a duplicate order.
- Separated payments into their own service, with the eSewa and Khalti gateways behind one interface.
- Handed orders to Nexus through a shared Redis instance, with REST as the fallback. Cached the catalogue in Redis.

Futsal matchmaking app, Android. Built in Kotlin. Players find venues and opponents by their preferences.